

5G RAN

CPRI Compression Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-04-27).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for CPRI compression by LampSite base stations in high frequency bands. For details, see: • 2.3 Differences • 5.1.1 CPRI Compression Process • 5.1.2.1 Determining the CPRI Compression Ratio Range	None	High-frequency TDD	DBS5900 LampSite

- Editorial changes

Revised descriptions of LampSite solutions. For details, see [5.1 Principles](#), [5.1.1 CPRI Compression Process](#), and [5.3.3 Hardware](#).

Revised descriptions of the requirements for configuring compression ratios over main and branch chains. For details, see [5.1.2.5 Determining the CPRI Compression Ratio](#).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-010017	CPRI Compression	4 CPRI Compression (Macro) 5 CPRI Compression (LampSite)

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
CPRI compression (macro)	None	4 CPRI Compression (Macro)
CPRI compression (LampSite)	None	5 CPRI Compression (LampSite)

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
CPRI compression (macro)	Supported in both high and low frequency bands, with the following differences: Only 2.5:1 and 3.2:1 CPRI compression are supported in high frequency bands. The compression ratios supported in low frequency bands vary depending on the RAT, cell bandwidth, and cell antenna TX/RX mode.	4 CPRI Compression (Macro)
CPRI compression (LampSite)		5 CPRI Compression (LampSite)

3 Overview

The Common Public Radio Interface (CPRI) protocol was established by the CPRI Cooperation. It defines an interface between a baseband unit (BBU) and a remote radio unit (RRU) or active antenna unit (AAU) in a macro base station, between a BBU and a radio HUB (RHUB) in a LampSite base station, or between an RHUB and a pico remote radio unit (pRRU) in a LampSite base station. After CPRI ports are connected, a link between them is established to transmit data. [Figure 3-1](#) shows CPRI-based data transmission for a macro base station, and [Figure 3-2](#) shows that for a LampSite base station.

Figure 3-1 CPRI-based data transmission for a macro base station (using RRUs as an example)

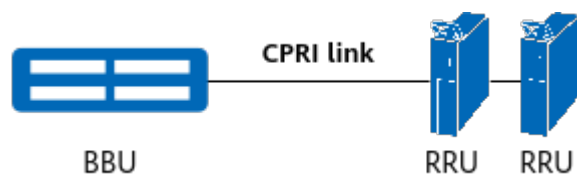


Figure 3-2 CPRI-based data transmission for a LampSite base station



Both control-plane and user-plane data are transmitted over the CPRI link as in-phase/quadrature (I/Q) data, which requires a significant amount of CPRI bandwidth resources in a cell. Due to the use of massive MIMO and high bandwidth, far more CPRI bandwidth resources are now required compared to GSM, UMTS, or LTE base stations. That is, an RRU/AAU/pRRU may need multiple pairs of optical modules and fibers to connect to the BBU/RHUB.

A number of factors influence the consumption of CPRI bandwidth resources, including the I/Q data sampling rate, the data bit width, and the number of antennas. The CPRI Compression feature compresses the I/Q data bit width and lowers the sampling rate to support the transmission of more I/Q data with the same CPRI bandwidth. This allows for more carriers or RRUs/AAUs/

RHUBs/pRRUs supported by a CPRI port, reducing the overall quantity and expenses of optical fibers, optical modules, and Ethernet cables.

In CPRI, the compression ratio refers to the ratio of consumed bandwidth resources when compression is disabled versus when it is enabled. The supported CPRI compression ratios vary depending on the cell configuration and hardware.

- For CPRI compression ratios supported by cells with different RATs, bandwidths, and TX/RX modes served by macro base stations, see [CPRI Compression Ratios Supported by Cells in Different Configurations](#). For CPRI compression ratios supported by such cells served by LampSite base stations, see [CPRI Compression Ratios of the Main and Branch Chains Supported by Cells in Different Configurations](#).
- For CPRI compression ratio requirements of hardware on macro base stations, see [4.3.3 Hardware](#). For CPRI compression ratio requirements of hardware on LampSite base stations, see [5.3.3 Hardware](#).

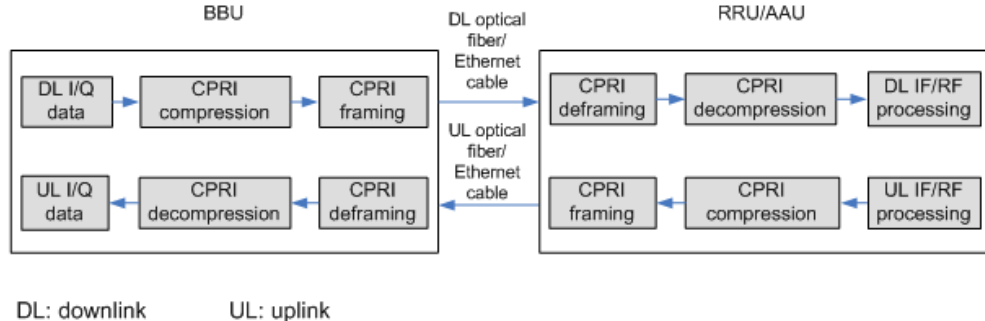
4 CPRI Compression (Macro)

4.1 Principles

4.1.1 CPRI Compression Process

Figure 4-1 shows the CPRI compression process.

Figure 4-1 CPRI compression process



- In the downlink
 - A BBU uses the CPRI compression algorithm to compress downlink I/Q data (user-plane data between a BBU and an RRU/AAU). The BBU transforms the data into CPRI frames and transmits these frames to the RRU/AAU through a downlink optical fiber or an Ethernet cable.
 - Upon receiving these frames, the RRU/AAU transforms them into compressed I/Q data, decompresses the data, and then transmits the decompressed I/Q data for further downlink intermediate frequency (IF)/radio frequency (RF) processing.
- In the uplink
 - An RRU/AAU uses the CPRI compression algorithm to compress uplink I/Q data from the IF/Rf module, transforms the data into CPRI frames, and transmits these frames to a BBU through an uplink optical fiber or an Ethernet cable.

- When the BBU receives these frames, it performs CPRI deframing and decompression.

 NOTE

CPRI compression (macro) applies only when the CPRI interface between the BBU and AAU/RRU uses the CPRI protocol. It does not apply to the CPRI interface using the eCPRI protocol. For details about the CPRI interface using the eCPRI protocol, see *eCPRI*.

4.1.2 CPRI Compression Ratio Selection Process

As a cell may support multiple CPRI compression ratios, the following procedures are required to select an appropriate one:

- [4.1.2.1 Determining the CPRI Compression Ratio Range](#)
- [4.1.2.2 Calculating the CPRI Payload Bandwidth over a CPRI Port](#)
- [4.1.2.3 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Disabled](#)
- [4.1.2.4 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Enabled](#)
- [4.1.2.5 Determining the CPRI Compression Ratio](#)

4.1.2.1 Determining the CPRI Compression Ratio Range

CPRI compression provides different compression ratios. The CPRI compression ratios supported by cells in different configurations and the corresponding 1T1R I/Q data bandwidths are listed in the tables under [CPRI Compression Ratios Supported by Cells in Different Configurations](#) and [1T1R I/Q Data Bandwidths](#), respectively. The **NRDUCellTrp.CpriCompression** parameter specifies the CPRI compression ratio:

- The value **NO_COMPRESSION** indicates that CPRI compression is not performed on the cell carrier.
- The value **1DOT5_COMPRESSION** indicates that 1.5:1 CPRI compression is performed on the cell carrier.
- The value **2_COMPRESSION** indicates that 2:1 CPRI compression is performed on the cell carrier.
- The value **2DOT2_COMPRESSION** indicates that 2.2:1 CPRI compression is performed on the cell carrier.
- The value **2DOT5_COMPRESSION** indicates that 2.5:1 CPRI compression is performed on the cell carrier.
- The value **3DOT2_COMPRESSION** indicates that 3.2:1 CPRI compression is performed on the cell carrier.

CPRI Compression Ratios Supported by Cells in Different Configurations

CPRI compression ratios supported by cells in different configurations are listed in [Table 4-1](#) and [Table 4-2](#). The compression ratios supported by an RF module can be queried based on its TX/RX mode and the cell bandwidth configuration. For example:

When a low-frequency NR TDD 4T4R RF module serves an NR TDD cell with a bandwidth of 100 MHz, the supported compression ratios are **3.2:1 CPRI Compression** and **2:1 CPRI Compression** according to [Table 4-1](#).

Table 4-1 CPRI compression ratios supported by TDD cells in different configurations

RAT	Cell Band width	TX/RX Mode	No CPRI Compression	1.5:1 CPRI Compression	2:1 CPRI Compression	2.2:1 CPRI Compression	2.5:1 CPRI Compression	3.2:1 CPRI Compression
Low-frequency TDD	20/30/40/50/60/70/80/90/100 MHz	1T1R/2T2R	No support	No support	Support	No support	No support	Support
		4T4R	No support	No support	Support	No support	No support	Support
		8T8R	No support	No support	Support	No support	No support	Support
		64T64R	No support	No support	No support	No support	No support	Support
High-frequency TDD	100/200 MHz	2T2R/4T4R	No support	No support	No support	No support	Support	Support

Note: For cells of the same type configured on one UBBPfw or high-frequency cells configured on one UBBPg, their CPRI compression ratios must be the same.

Table 4-2 CPRI compression ratios supported by SUL cells in different configurations

RAT	Cell Band width	TX/RX Mode	No CPRI Compression	1.5:1 CPRI Compression	2:1 CPRI Compression	2.2:1 CPRI Compression	2.5:1 CPRI Compression	3.2:1 CPRI Compression
SUL	10 MHz	2R/4R	Support	Support	No support	No support	No support	No support
SUL	15 MHz	2R/4R	Support	No support	Support	No support	No support	No support

RAT	Cell Band width	TX/RX Mode	No CPRI Compression	1.5:1 CPRI Compression	2:1 CPRI Compression	2.2:1 CPRI Compression	2.5:1 CPRI Compression	3.2:1 CPRI Compression
SUL	20 MHz	2R/4R	Support	No support	Support	No support	No support	No support

1T1R I/Q Data Bandwidths

The 1T1R I/Q data bandwidths in cells with different configurations are listed in [Table 4-1](#) and [Table 4-2](#).

Table 4-3 1T1R I/Q data bandwidths in TDD cells with different configurations

RAT	Cell Bandwidth	No CPRI Compression (Gbit/s)	1.5:1 CPRI Compression (Gbit/s)	2:1 CPRI Compression (Gbit/s)	2.2:1 CPRI Compression (Gbit/s)	2.5:1 CPRI Compression (Gbit/s)	3.2:1 CPRI Compression (Gbit/s)
Low-frequency TDD	20 MHz	N/A	N/A	0.4608	N/A	N/A	0.2872
Low-frequency TDD	30/40 MHz	N/A	N/A	0.9216	N/A	N/A	0.5745
Low-frequency TDD	50/60 MHz	N/A	N/A	1.3824	N/A	N/A	0.8617
Low-frequency TDD	70/80 MHz	N/A	N/A	1.8432	N/A	N/A	1.1489
Low-frequency TDD	90/100 MHz	N/A	N/A	2.304	N/A	N/A	1.4362
High-frequency TDD	100 MHz	N/A	N/A	N/A	N/A	1.8432	1.4362
High-frequency TDD	200 MHz	N/A	N/A	N/A	N/A	3.6864	2.8724

RAT	Cell Bandwidth	No CPRI Compression (Gbit/s)	1.5:1 CPRI Compression (Gbit/s)	2:1 CPRI Compression (Gbit/s)	2.2:1 CPRI Compression (Gbit/s)	2.5:1 CPRI Compression (Gbit/s)	3.2:1 CPRI Compression (Gbit/s)
Note: "N/A" in the table indicates that the corresponding CPRI compression ratio is not supported in a certain bandwidth.							

Table 4-4 1T1R I/Q data bandwidths in SUL cells with different configurations

RAT	Cell Bandwidth	No CPRI Compression (Gbit/s)	1.5:1 CPRI Compression (Gbit/s)	2:1 CPRI Compression (Gbit/s)	2.2:1 CPRI Compression (Gbit/s)	2.5:1 CPRI Compression (Gbit/s)	3.2:1 CPRI Compression (Gbit/s)
SUL	10 MHz	0.4608	0.3072	N/A	N/A	N/A	N/A
SUL	15 MHz	0.9216	N/A	0.4608	N/A	N/A	N/A
SUL	20 MHz	0.9216	N/A	0.4608	N/A	N/A	N/A
Note: "N/A" in the table indicates that the corresponding CPRI compression ratio is not supported in a certain bandwidth.							

4.1.2.2 Calculating the CPRI Payload Bandwidth over a CPRI Port

CPRI payload bandwidth is used to transmit I/Q data over a CPRI port. It is calculated using the following formula: CPRI payload bandwidth over a CPRI port = CPRI bandwidth x (1 – Control data overhead) x Coding efficiency.

Specifically,

- **CPRI bandwidth**

The CPRI bandwidth equals the negotiated CPRI line rate. Huawei supports the following CPRI line rates defined in section 4.2.1 "Line Bit Rate" of CPRI Specification V7.0: 2.4576 Gbit/s, 4.9152 Gbit/s, 6.144 Gbit/s, 9.8304 Gbit/s, 10.1376 Gbit/s, 24.33024 Gbit/s (25G for short), and 4 x 24.33024 Gbit/s (100G for short).

- **Control data overhead**

CPRI specifications define a part of a given bandwidth for control-plane transmission. The proportion of control-plane data varies with the CPRI bandwidth. For details, see section 4.2.7 "Frame Structure" of CPRI Specification V7.0.

- **Coding efficiency**

The coding efficiency refers to the forward error correction (FEC) coding efficiency at the physical layer. As defined in section 4.2.1 "Line Bit Rate" of

CPRI Specification V7.0, the coding efficiency can be 8B/10B (8/10 for short) or 64B/66B (64/66 for short).

If the CPRI bandwidth is 2.4576 Gbit/s, then: CPRI payload bandwidth = $2.4576 \times (1 - 1/16) \times 8/10 = 1.8432$ Gbit/s. [Table 4-5](#) lists the control data overhead, coding efficiency, and calculated CPRI payload bandwidth corresponding to different CPRI bandwidths.

Table 4-5 Control data overhead, coding efficiency, and calculated CPRI payload bandwidth corresponding to different CPRI bandwidths

CPRI Bandwidth	Control Data Overhead	Coding Efficiency	CPRI Payload Bandwidth
2.4576 Gbit/s	1/16	8/10	1.8432 Gbit/s
4.9152 Gbit/s	1/16	8/10	3.6864 Gbit/s
6.144 Gbit/s	1/16	8/10	4.608 Gbit/s
9.8304 Gbit/s	1/16	8/10	7.3728 Gbit/s
10.1376 Gbit/s	1/20	64/66	9.3389 Gbit/s
24.33024 Gbit/s (25G for short)	1/48	64/66	23.10144 Gbit/s
4 x 24.33024 Gbit/s (100G for short)	1/48	64/66	4 x 23.10144 Gbit/s

 NOTE

24.33024 Gbit/s is a CPRI line rate defined in CPRI specifications. If a 25 Gbit/s SFP optical module is used, the CPRI port supports a maximum of 24.33024 Gbit/s bandwidth. If a 100 Gbit/s QSFP optical module is used, the CPRI port supports a maximum of 4 x 24.33024 Gbit/s.

4.1.2.3 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Disabled

The I/Q data bandwidth required between a BBU and an RRU or AAU when CPRI compression is disabled is calculated using the following formula:

I/Q data bandwidth required when CPRI compression is disabled = Number of antennas in cell 1 x 1T1R I/Q data bandwidth + Number of antennas in cell 2 x 1T1R I/Q data bandwidth + ... + Number of antennas in cell m x 1T1R I/Q data bandwidth

Specifically,

- The number of antennas is determined by the onsite antenna configuration for a cell. If the numbers of TX and RX antennas are different, the larger value is used. For example, the number of antennas is 4 for a 4T4R cell as well as a 2T4R cell.

- The 1T1R I/Q data bandwidth depends on the cell bandwidth. For details about the 1T1R I/Q data bandwidths when different cell bandwidths are used, see the **No CPRI Compression (Gbit/s)** column of the tables in [1T1R I/Q Data Bandwidths](#).
- m indicates the total number of cells served by a single CPRI link.

4.1.2.4 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Enabled

The I/Q data bandwidth required between a BBU and an RRU or AAU when CPRI compression is enabled is calculated using the following formula:

I/Q data bandwidth required when CPRI compression is enabled = Number of antennas in cell 1 x 1T1R I/Q data bandwidth + Number of antennas in cell 2 x 1T1R I/Q data bandwidth + ... + Number of antennas in cell m x 1T1R I/Q data bandwidth

Specifically,

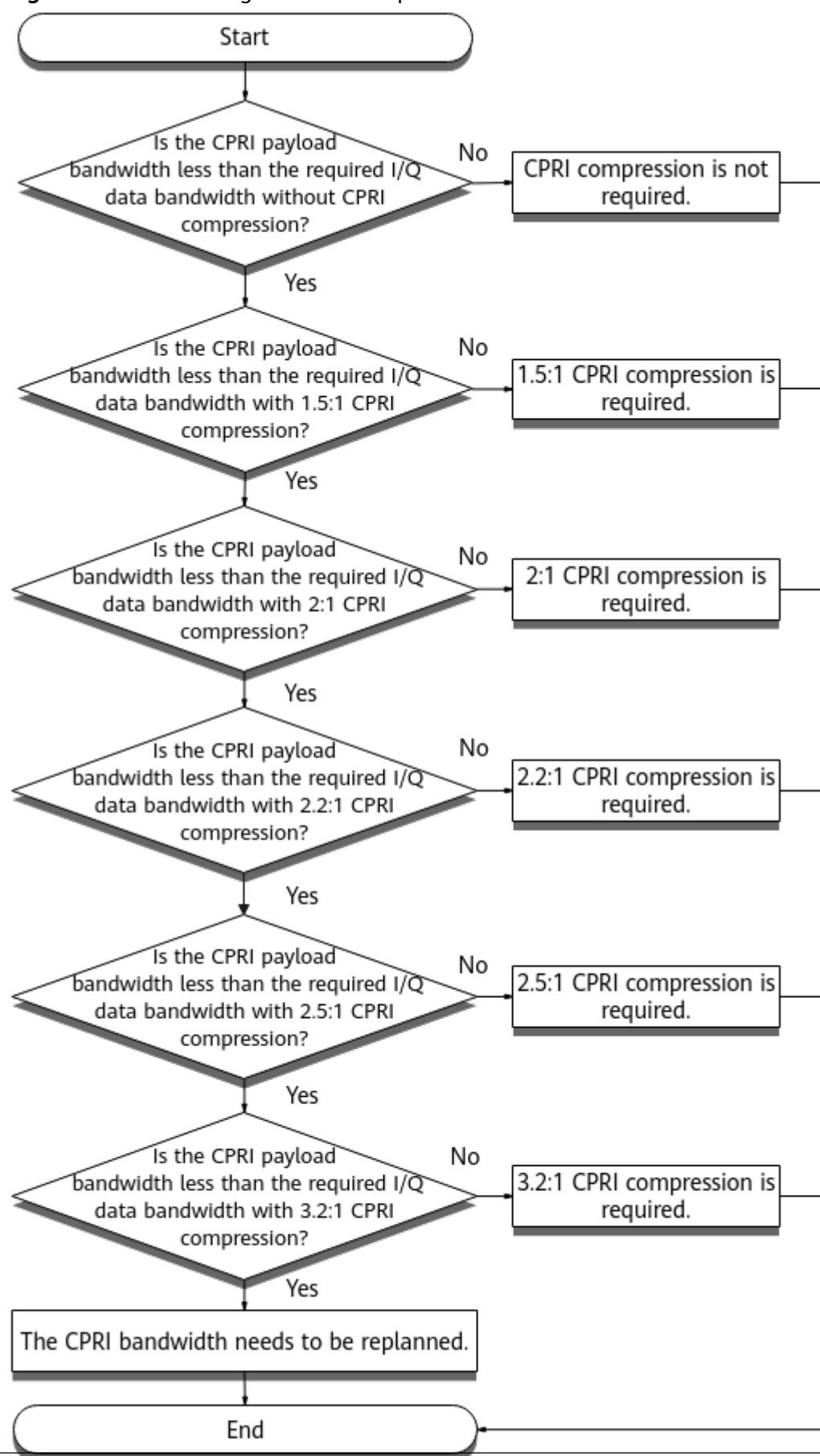
- The number of antennas is determined by the onsite antenna configuration for a cell. If the numbers of TX and RX antennas are different, the larger value is used. For example, the number of antennas is 4 for a 4T4R cell as well as a 2T4R cell.
- The 1T1R I/Q data bandwidth depends on the cell bandwidth. For details about the 1T1R I/Q data bandwidths when different cell bandwidths are used, see the **1.5:1 CPRI Compression (Gbit/s)**, **2:1 CPRI Compression (Gbit/s)**, **2.5:1 CPRI Compression (Gbit/s)**, or **3.2:1 CPRI Compression (Gbit/s)** column of the tables in [1T1R I/Q Data Bandwidths](#).
- m indicates the total number of cells served by a single CPRI link.

The 1T1R I/Q data bandwidth decreases after CPRI compression is enabled, and likewise, the required I/Q data bandwidth also decreases.

4.1.2.5 Determining the CPRI Compression Ratio

The CPRI compression ratios supported by a cell are determined based on its configurations (as listed in [CPRI Compression Ratios Supported by Cells in Different Configurations](#)), after which an appropriate CPRI compression ratio is selected according to the procedure shown in [Figure 4-2](#).

Figure 4-2 Determining the CPRI compression ratio



4.1.3 Reporting an Alarm Indicating Insufficient eCPRI/CPRI Resources

To solve the problem that no alarm is reported when eCPRI/CPRI resources are insufficient in a cell, the function of reporting an alarm indicating insufficient eCPRI/CPRI resources is introduced. The function is recommended in 8T8R or higher scenarios and is controlled by the **ECPRI_CPRI_INSUFT_ALM_RPT_SW** option of the **gNBOamParam.AlarmSwitch** parameter. If this function is enabled, ALM-29871 NR DU Cell TRP Capability Degraded is reported with **Specific Problem** being **Insufficient CPRI bandwidth** in the scenario where the eCPRI bandwidth does not meet the maximum service requirement or 1/8 or fewer channels have not been allocated due to insufficient cell CPRI bandwidth or backplane bandwidth. If this function is disabled, this alarm is not reported in the preceding scenario.

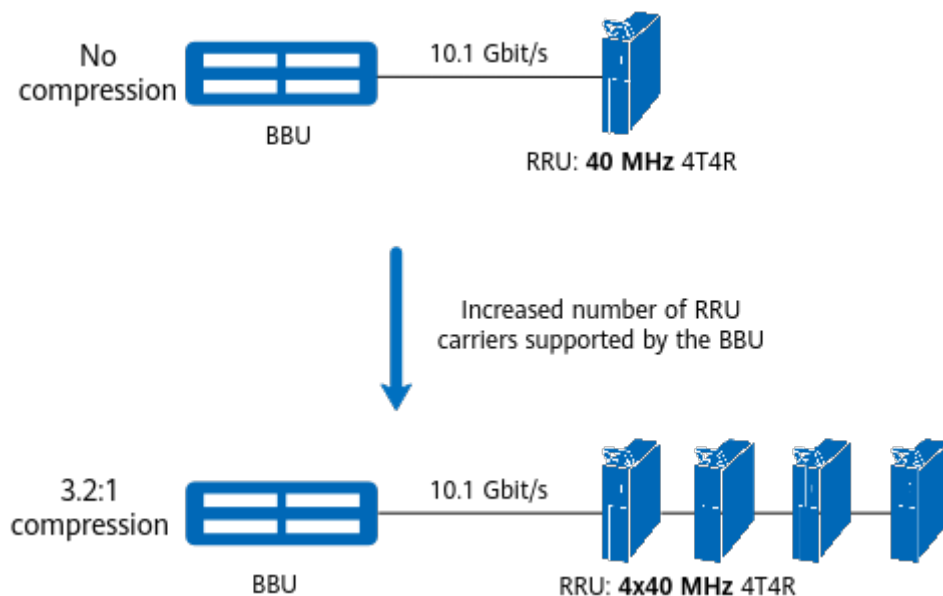
4.2 Network Analysis

4.2.1 Benefits

When CPRI bandwidth is insufficient, CPRI compression can be applied to provide the following benefits (using RRUs and a 3.2:1 CPRI compression ratio as an example):

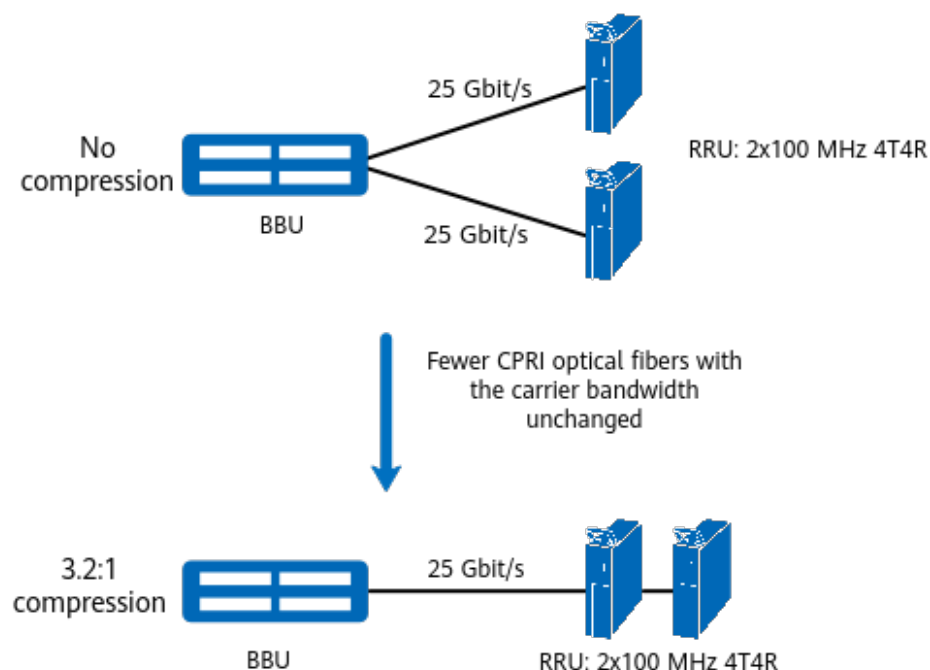
- Increases the number of RRU carriers supported by the BBU, as shown in [Figure 4-3](#).

Figure 4-3 Increased number of RRU carriers supported by the BBU



- Reduces the number of CPRI optical fibers and costs, as shown in [Figure 4-4](#).

Figure 4-4 Reduced number of CPRI optical fibers



4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

CPRI compression impacts network performance in the following ways:

- In the uplink, the signal to interference plus noise ratio (SINR) decreases slightly when interference is strong.
- In the downlink, the error vector magnitude (EVM) of the base station slightly increases, and the SINR slightly decreases.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RRUs and CPRI-compliant NR-capable AAUs support this function. For details, see the technical specifications of RRUs or AAUs in *3900 & 5900 Series Base Station Product Documentation*.

In particular,

- When an RRU5254, RRU5258, or RRU5836E serves an NR cell with 3.2:1 CPRI compression:
 - LTE cells with CPRI compression enabled cannot be configured on this RF module.
 - NR cells with other compression ratios cannot be configured on this RF module.
- The compression ratios supported by different RF modules can be queried in [CPRI Compression Ratios Supported by Cells in Different Configurations](#) based on their TX/RX modes and the cell bandwidth configuration.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

The configuration of a CPRI compression ratio must be performed based on CPRI compression ratios supported by cells in different configurations (as listed in [CPRI Compression Ratios Supported by Cells in Different Configurations](#)). Otherwise, the configuration may fail to take effect.

Table 4-6 describes the parameters used for function activation.

Table 4-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
TRP Type	NRDUCellTrp.TrpType	Set this parameter based on the network plan.
CPRI Compression	NRDUCellTrp.CpriCompression	• To enable 3.2:1 CPRI compression, set this parameter to 3DOT2_COMPRESSION. • To enable 2.5:1 CPRI compression, set this parameter to 2DOT5_COMPRESSION. • To enable 2.2:1 CPRI compression, set this parameter to 2DOT2_COMPRESSION. • To enable 2:1 CPRI compression, set this parameter to 2_COMPRESSION. • To enable 1.5:1 CPRI compression, set this parameter to 1DOT5_COMPRESSION. • To disable CPRI compression, set this parameter to NO_COMPRESSION.

NOTICE

- The CPRI compression ratio cannot be modified when a cell is active. The cell must be deactivated before changing the CPRI compression ratio.
- In high-frequency scenarios, cells over a CPRI link must use the same CPRI compression ratio. Mixed configuration is not supported.
- If an RF module does not support the configured CPRI compression ratio, cells on the RF module cannot be activated. For details about the CPRI compression ratios supported by RF modules, see [CPRI Compression Ratios Supported by Cells in Different Configurations](#) and [RF Modules](#).

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring CPRI compression (taking the 3.2:1 CPRI compression ratio as an example)  
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=3DOT2_COMPRESSION;
```

Deactivation Command Examples

```
//Deactivating CPRI compression  
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=NO_COMPRESSION;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Use either of the following methods to check whether CPRI compression has been activated and taken effect:

Method 1:

The following uses the 3.2:1 compression ratio as an example:

- Step 1** Run the **LST NRDUCELLTRP** command to verify that the value of **CPRI Compression** is **3.2:1 Compression**.
- Step 2** Run the **DSP NRCELL** command to verify that the value of **Cell Latest State Change Reason** is **NR Cell Setup Success**.

----End

Method 2:

Run the **DSP NRDUCELLTRPCPRIINFO** command. If the value of **Total Bandwidth of Cell CPRI Link(Mbit/s)** is less than that when CPRI compression is disabled, CPRI compression has taken effect.

4.4.3 Network Monitoring

If CPRI compression is enabled and the **ECPRI_CPRI_INSUFT_ALM_RPT_SW** option of the **gNBOamParam.AlarmSwitch** parameter is selected, ALM-29871 NR DU

Cell TRP Capability Degraded will be reported with **Specific Problem** being **Insufficient CPRI bandwidth** in the scenario where 1/8 or fewer channels have not been allocated due to insufficient cell CPRI bandwidth or backplane bandwidth. In this case, the CPRI bandwidth needs to be replanned.

5 CPRI Compression (LampSite)

5.1 Principles

The network architecture of each LampSite solution involves both the main and branch chains. This section describes only CPRI compression in the BBU access scenarios of LampSite solutions. For details on CPRI compression ratio selection in the RF input scenarios, contact Huawei engineers.

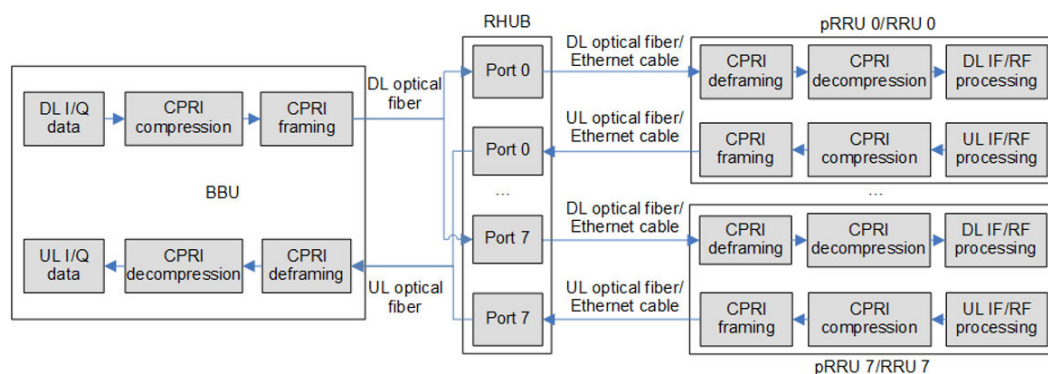
NOTE

For details on LampSite solutions, see *LampSite in 3900 & 5900 Series Base Station Product Documentation*.

5.1.1 CPRI Compression Process

Figure 5-1 and **Figure 5-2** show the CPRI compression processes when non-next-generation and next-generation LampSite products are used, respectively.

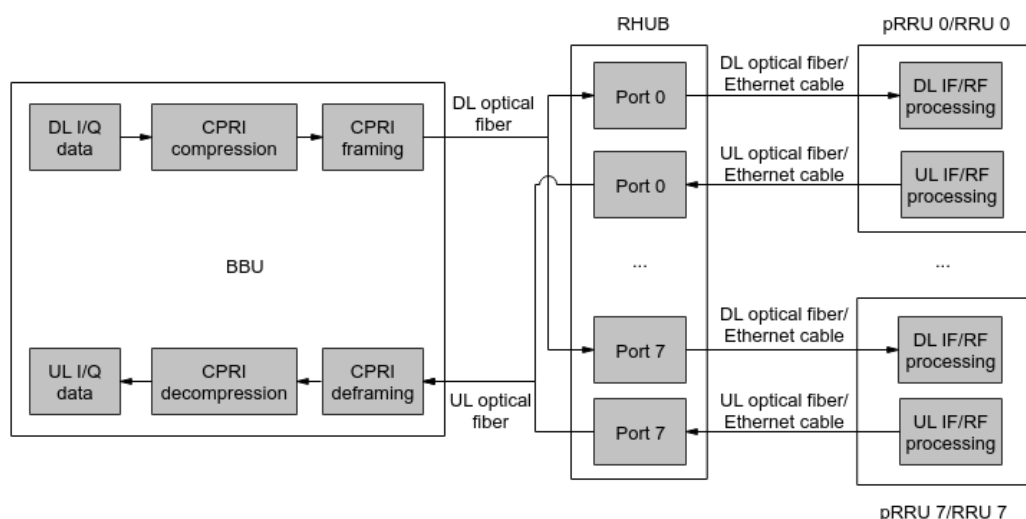
Figure 5-1 CPRI compression process when non-next-generation LampSite products are used



- In the downlink
 - Main chain: A BBU uses the CPRI compression algorithm to compress downlink I/Q data. It then transforms the data into CPRI frames and transmits these frames to an RHUB through a downlink optical fiber.

- Branch chain: An RHUB compresses the received CPRI frames and then sends them to a pRRU over a downlink Ethernet cable or optical fiber. When the pRRU receives the CPRI frames, it then performs CPRI deframing and decompression, and then transmits the decompressed I/Q data to the IF/RF module.
- In the uplink
 - Branch chain: A pRRU uses the CPRI compression algorithm to compress the I/Q data received from the IF/RF module, transforms the data into CPRI frames, and transmits these frames to an RHUB through an uplink Ethernet cable or optical fiber.
 - Main chain: An RHUB compresses the received CPRI frames and then sends them to a BBU over an uplink optical fiber. After receiving the CPRI frames, the BBU performs CPRI deframing and decompression.

Figure 5-2 CPRI compression process when next-generation LampSite products are used



- In the downlink
 - Main chain: A BBU uses the CPRI compression algorithm to compress downlink I/Q data. It then transforms the data into CPRI frames and transmits these frames to an RHUB through a downlink optical fiber.
 - Branch chain: An RHUB transmits the antenna data to a pRRU through a downlink Ethernet cable or optical fiber. After receiving the data, the pRRU transmits it to the IF/RF module for processing.
- In the uplink
 - Branch chain: A pRRU transmits the data received from the IF/RF module to an RHUB through an uplink Ethernet cable or optical fiber.
 - Main chain: An RHUB compresses the received data and then sends them to a BBU over an uplink optical fiber. After receiving the data, the BBU performs CPRI deframing and decompression.

 NOTE

Next-generation LampSite products include pRRU555xAB, pRRU565xAB, pRRU575xAB, and pRRU585xAB modules as well as their corresponding RHUBs. *x* is a digit; *A* and *B* stand for different letters. For details about the mapping between pRRUs and RHUBs, see the pRRU or RHUB technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

LampSite products of other models do not belong to the next-generation category.

5.1.2 CPRI Compression Ratio Selection Process

As multiple CPRI compression ratios are available for the main and branch chains of a cell, the following procedures are required to select appropriate ratios:

- [5.1.2.1 Determining the CPRI Compression Ratio Range](#)
- [5.1.2.2 Calculating the CPRI Payload Bandwidth over a CPRI Port](#)
- [5.1.2.3 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Disabled](#)
- [5.1.2.4 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Enabled](#)
- [5.1.2.5 Determining the CPRI Compression Ratio](#)

5.1.2.1 Determining the CPRI Compression Ratio Range

CPRI compression provides different compression ratios. The CPRI compression ratios of the main and branch chains supported by cells in different configurations and the corresponding 1T1R data bandwidths are listed in the tables under [CPRI Compression Ratios of the Main and Branch Chains Supported by Cells in Different Configurations](#), [1T1R I/Q Data Bandwidth over the Main Chain](#), and [1T1R I/Q Data Bandwidth over the Branch Chain](#). The CPRI compression ratio of the main chain is specified by **NRDUCellTrp.CpriCompression**. When non-next-generation LampSite products are used, the CPRI compression ratio of a branch chain is specified by **NRDUCellTrp.BranchCpriCompression**. The values of these parameters are described as follows:

- The value **NO_COMPRESSION** indicates that CPRI compression is not performed on the cell carrier.
- The value **1DOT5_COMPRESSION** indicates that 1.5:1 CPRI compression is performed on the cell carrier.
- The value **2_COMPRESSION** indicates that 2:1 CPRI compression is performed on the cell carrier.
- The value **2DOT5_COMPRESSION** indicates that 2.5:1 CPRI compression is performed on the cell carrier.
- The value **3_COMPRESSION** indicates that 3:1 CPRI compression is performed on the cell carrier.
- The value **3DOT2_COMPRESSION** indicates that 3.2:1 CPRI compression is performed on the cell carrier.
- The value **4_COMPRESSION** indicates that 4:1 CPRI compression is performed on the cell carrier.

For next-generation LampSite products, there is no limit on bandwidths because a new fronthaul technology is used. Therefore, CPRI compression is not involved

over branch chains, and **NRDUCellTrp.BranchCpriCompression** does not take effect even after being configured.

CPRI Compression Ratios of the Main and Branch Chains Supported by Cells in Different Configurations

CPRI compression ratios supported by cells in different configurations are listed in [Table 5-1](#).

Table 5-1 CPRI compression ratios supported by TDD cells in different configurations

RA T	Cell Ban dwi dth	TX/ RX Mod e	No CPRI Compr ession	1.5:1 CPRI Comp ressio n	2:1 CPRI Comp ressio n	2.5:1 CPRI Comp ressio n	3:1 CPRI Comp ressio n	3.2:1 CPRI Comp ressio n	4:1 CPRI Comp ressio n
Lo w- freq uen cy TD D	20 MHz	2T2 R/ 4T4 R	No suppor t	No suppor t	No suppor t	No suppor t	No suppor t	Suppo rt	No suppor t
Lo w- freq uen cy TD D	30/4 0 MHz	2T2 R/ 4T4 R	No suppor t	No suppor t	No suppor t	No suppor t	No suppor t	Suppo rt	No suppor t
Lo w- freq uen cy TD D	50/6 0 MHz	2T2 R/ 4T4 R	No suppor t	No suppor t	No suppor t	No suppor t	No suppor t	Suppo rt	No suppor t
Lo w- freq uen cy TD D	70/8 0 MHz	2T2 R/ 4T4 R	No suppor t	No suppor t	No suppor t	No suppor t	No suppor t	Suppo rt	No suppor t

RA T	Cell Ban dwi dth	TX/ RX Mod e	No CPRI Compr ession	1.5:1 CPRI Comp ressio n	2:1 CPRI Comp ressio n	2.5:1 CPRI Comp ressio n	3:1 CPRI Comp ressio n	3.2:1 CPRI Comp ressio n	4:1 CPRI Comp ressio n
Lo w- freq uen cy TD D	90/1 00 MHz	2T2 R/ 4T4 R	No suppor t	No suppo rt	No suppo rt	No suppo rt	No suppo rt	Suppo rt	No suppo rt
Hig h- freq uen cy TD D	100 MHz	2T2 R	No suppor t	No suppo rt	No suppo rt	Suppo rt	No suppo rt	Suppo rt	No suppo rt
Hig h- freq uen cy TD D	200 MHz	2T2 R	No suppor t	No suppo rt	No suppo rt	Suppo rt	No suppo rt	Suppo rt	No suppo rt
Note 1: In the table, "Support" or "No support" indicates that both the main and branch chains support CPRI compression or neither of them supports CPRI compression. Note 2: For cells of the same type configured on one UBBPfw, their CPRI compression ratios must be the same.									

1T1R I/Q Data Bandwidth over the Main Chain

1T1R I/Q data bandwidths over the main chain in cells with different configurations are listed in [Table 5-2](#).

Table 5-2 1T1R I/Q data bandwidths over the main chain in TDD cells with different configurations

RAT	Cell Bandwidth	No CPRI Compression (Gbit/s)	1.5:1 CPRI Compression (Gbit/s)	2:1 CPRI Compression (Gbit/s)	2.5:1 CPRI Compression (Gbit/s)	3.2:1 CPRI Compression (Gbit/s)
Low-frequency TDD	20 MHz	N/A	N/A	N/A	N/A	0.2872
Low-frequency TDD	30/40 MHz	N/A	N/A	N/A	N/A	0.5745
Low-frequency TDD	50/60 MHz	N/A	N/A	N/A	N/A	0.8617
Low-frequency TDD	70/80 MHz	N/A	N/A	N/A	N/A	1.1489
Low-frequency TDD	90/100 MHz	N/A	N/A	N/A	N/A	1.4362
High-frequency TDD	100 MHz	N/A	N/A	N/A	1.8432	1.4362
High-frequency TDD	200 MHz	N/A	N/A	N/A	3.6864	2.8724
Note: In this table, "N/A" indicates that the 1T1R I/Q data bandwidth with a compression ratio is not applicable for a certain bandwidth.						

1T1R I/Q Data Bandwidth over the Branch Chain

1T1R I/Q data bandwidths over the branch chain in cells with different configurations are listed in [Table 5-3](#).

Table 5-3 1T1R I/Q data bandwidths over the branch chain in TDD cells with different configurations

RAT	Cell Bandwidth	2:1 CPRI Compression (Gbit/s)	2.5:1 CPRI Compression (Gbit/s)	3:1 CPRI Compression (Gbit/s)	3.2:1 CPRI Compression (Gbit/s)	4:1 CPRI Compression (Gbit/s)
Low-frequency TDD	20 MHz	N/A	N/A	N/A	0.2872	N/A
Low-frequency TDD	30/40 MHz	N/A	N/A	N/A	0.5745	N/A
Low-frequency TDD	50/60 MHz	N/A	N/A	N/A	0.8617	N/A
Low-frequency TDD	70/80 MHz	N/A	N/A	N/A	1.1489	N/A
Low-frequency TDD	90/100 MHz	N/A	N/A	N/A	1.4362	N/A
High-frequency TDD	100 MHz	N/A	1.8432	N/A	1.4362	N/A

RAT	Cell Bandwidth	2:1 CPRI Compression (Gbit/s)	2.5:1 CPRI Compression (Gbit/s)	3:1 CPRI Compression (Gbit/s)	3.2:1 CPRI Compression (Gbit/s)	4:1 CPRI Compression (Gbit/s)
High - frequency TDD	200 MHz	N/A	3.6864	N/A	2.8724	N/A
Note: In this table, "N/A" indicates that the 1T1R I/Q data bandwidth with a compression ratio is not applicable for a certain bandwidth.						

5.1.2.2 Calculating the CPRI Payload Bandwidth over a CPRI Port

CPRI payload bandwidth is used to transmit I/Q data over a CPRI port. It is calculated using the following formula: CPRI payload bandwidth over a CPRI port = CPRI bandwidth x (1 – Control data overhead) x Coding efficiency.

Specifically,

- **CPRI bandwidth (main chain)**

The CPRI bandwidth equals the negotiated CPRI line rate. Huawei supports the following CPRI line rates defined in section 4.2.1 "Line Bit Rate" of CPRI Specification V7.0: 4.9152 Gbit/s, 9.8304 Gbit/s, 10.1376 Gbit/s, and 24.33024 Gbit/s (25G for short).

- **CPRI bandwidth (branch chain)**

The CPRI bandwidth equals the configured CPRI line rate. Huawei supports the following CPRI line rates defined in section 4.2.1 "Line Bit Rate" of CPRI Specification V7.0: 1.2288 Gbit/s, 3.072 Gbit/s, 10.1376 Gbit/s, and 24.33024 Gbit/s.

- **Control data overhead**

CPRI specifications define a part of a given bandwidth for control-plane transmission. The proportion of control-plane data varies with the CPRI bandwidth. For details, see section 4.2.7 "Frame Structure" of CPRI Specification V7.0.

- **Coding efficiency**

The coding efficiency refers to the FEC coding efficiency at the physical layer. As defined in section 4.2.1 "Line Bit Rate" of CPRI Specification V7.0, the coding efficiency can be 8B/10B (8/10 for short) or 64B/66B (64/66 for short).

If the CPRI bandwidth of a branch chain is 1.2288 Gbit/s, then: CPRI payload bandwidth = 1.2288 x (1 – 1/16) x 8/10 = 0.9216 Gbit/s. [Table 5-4](#) lists the control data overhead, coding efficiency, and calculated CPRI payload bandwidth corresponding to different CPRI bandwidths.

Table 5-4 Control data overhead, coding efficiency, and calculated CPRI payload bandwidth corresponding to different CPRI bandwidths

CPRI Bandwidth	Control Data Overhead	Coding Efficiency	CPRI Payload Bandwidth
1.2288 Gbit/s	1/16	8/10	0.9216 Gbit/s
3.072 Gbit/s	1/16	8/10	2.304 Gbit/s
4.9152 Gbit/s	1/16	8/10	3.6864 Gbit/s
9.8304 Gbit/s	1/16	8/10	7.3728 Gbit/s
10.1376 Gbit/s	1/20	64/66	9.3389 Gbit/s
24.33024 Gbit/s (25G for short)	1/48	64/66	23.10144 Gbit/s

5.1.2.3 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Disabled

The I/Q data bandwidth required between a BBU and an RHUB or between an RHUB and a pRRU when CPRI compression is disabled is calculated using the following formula:

I/Q data bandwidth required when CPRI compression is disabled = Number of antennas in cell 1 x 1T1R I/Q data bandwidth + Number of antennas in cell 2 x 1T1R I/Q data bandwidth + ... + Number of antennas in cell m x 1T1R I/Q data bandwidth

Specifically,

- The number of antennas is determined by the onsite antenna configuration for a cell. For example, the number of antennas is 4 for a 4T4R cell.
- The 1T1R I/Q data bandwidth depends on the cell bandwidth. For details about the 1T1R I/Q data bandwidths when different cell bandwidths are used, see the **No CPRI Compression (Gbit/s)** column of the tables in [1T1R I/Q Data Bandwidth over the Main Chain](#) and [1T1R I/Q Data Bandwidth over the Branch Chain](#).
- m indicates the total number of cells served by a single CPRI link.

5.1.2.4 Calculating the I/Q Data Bandwidth Required When CPRI Compression Is Enabled

The I/Q data bandwidth required between a BBU and an RHUB or between an RHUB and a pRRU when CPRI compression is enabled is calculated using the following formula:

I/Q data bandwidth required when CPRI compression is enabled = Number of antennas in cell 1 x 1T1R I/Q data bandwidth + Number of antennas in cell 2 x 1T1R I/Q data bandwidth + ... + Number of antennas in cell m x 1T1R I/Q data bandwidth

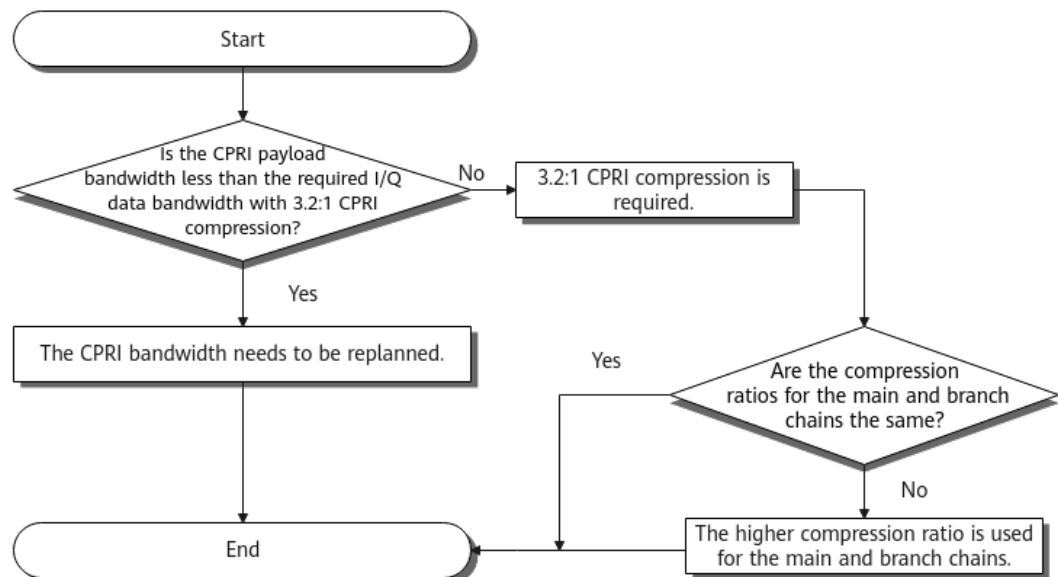
Specifically,

- The number of antennas is determined by the onsite antenna configuration for a cell. For example, the number of antennas is 4 for a 4T4R cell.
- The 1T1R I/Q data bandwidth depends on the cell bandwidth. For details about the 1T1R I/Q data bandwidths when different cell bandwidths are used, see the **1.5:1 CPRI Compression (Gbit/s)**, **2:1 CPRI Compression (Gbit/s)**, or **3.2:1 CPRI Compression (Gbit/s)** column of the tables in **1T1R I/Q Data Bandwidth over the Main Chain** and **1T1R I/Q Data Bandwidth over the Branch Chain**.
- m indicates the total number of cells served by a single CPRI link.

5.1.2.5 Determining the CPRI Compression Ratio

The CPRI compression ratios used by NR TDD main and branch chains can be selected according to the procedure in [Figure 5-3](#), and these ratios must be the same.

Figure 5-3 CPRI compression ratio selection in NR TDD



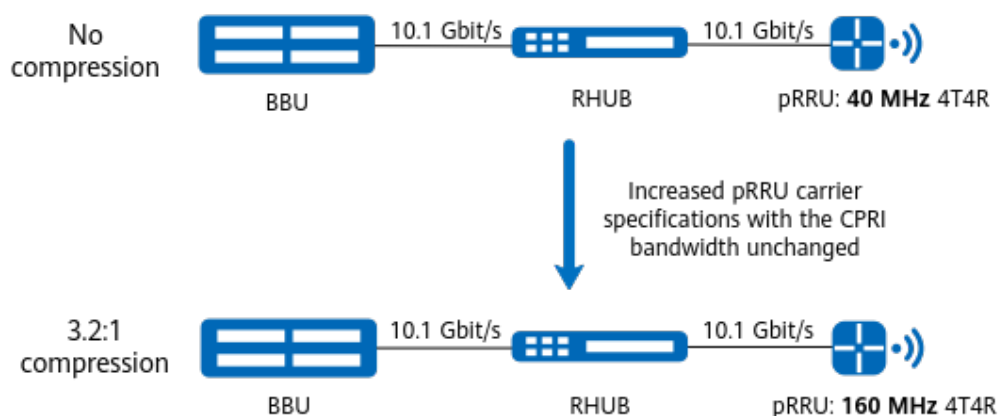
5.2 Network Analysis

5.2.1 Benefits

When the CPRI bandwidth is insufficient, CPRI compression is applied to provide the following benefits (using 3.2:1 CPRI compression as an example):

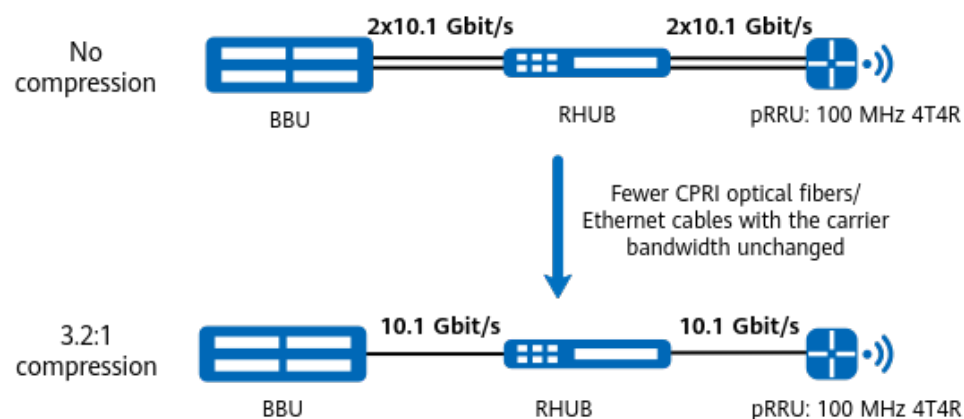
- Increases the pRRU carrier specifications, as shown in [Figure 5-4](#).

Figure 5-4 Increased pRRU carrier specifications



- Reduces the number of CPRI optical fibers/Ethernet cables and costs, as shown in [Figure 5-5](#).

Figure 5-5 Reduced number of CPRI optical fibers/Ethernet cables



5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

CPRI compression impacts network performance in the following ways:

- In the uplink, the SINR decreases slightly when interference is strong.
- In the downlink, the EVM of the base station slightly increases, and the SINR slightly decreases.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable pRRUs support this function. For details about the pRRUs supporting LampSite solutions, see sections "Configuration Principles" in "LampSite Solution Description" of *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-5](#) describes the parameters used for function activation.

Table 5-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
TRP Type	NRDUCellTrp.TrpType	Set this parameter based on the network plan.
CPRI Compression	NRDUCellTrp.CpriCompression	• To enable 3.2:1 CPRI compression, set this parameter to 3DOT2_COMPRESSION. • To enable 2.5:1 CPRI compression, set this parameter to 2DOT5_COMPRESSION. • To enable 2:1 CPRI compression, set this parameter to 2_COMPRESSION. • To enable 1.5:1 CPRI compression, set this parameter to 1DOT5_COMPRESSION. • To disable CPRI compression, set this parameter to NO_COMPRESSION.
Branch CPRI Compression	NRDUCellTrp.BranchCpriCompression	• To enable 4:1 CPRI compression, set this parameter to 4_COMPRESSION. • To enable 3.2:1 CPRI compression, set this parameter to 3DOT2_COMPRESSION. • To enable 3:1 CPRI compression, set this parameter to 3_COMPRESSION. • To enable 2.5:1 CPRI compression, set this parameter to 2DOT5_COMPRESSION. • To enable 2:1 CPRI compression, set this parameter to 2_COMPRESSION. • To disable CPRI compression, set this parameter to NO_COMPRESSION.

NOTICE

The CPRI compression ratio cannot be modified when a cell is active. The cell must be deactivated before changing the CPRI compression ratio.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring 3.2:1 CPRI compression for the main chain
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, CpriCompression=3DOT2_COMPRESSION;
//Configuring 3.2:1 CPRI compression for the branch chain (when non-next-generation LampSite products
are used)
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=DEFAULT, BranchCpriCompression=3DOT2_COMPRESSION;
```

Deactivation Command Examples

```
//Deactivating CPRI compression for the main chain
MOD NRDUCELLTRP: NrDuCellTrpId =0, TrpType=DEFAULT, CpriCompression=NO_COMPRESSION;
//Deactivating CPRI compression for the branch chain (when non-next-generation LampSite products are
used)
MOD NRDUCELLTRP: NrDuCellTrpId =0, TrpType=DEFAULT, BranchCpriCompression=NO_COMPRESSION;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

- Perform the following steps to check whether CPRI compression has been activated and taken effect on the main chain:

Step 1 Run the **LST NRDUCELLTRP** command to verify that the value of **CPRI Compression** is **3.2:1 Compression**.

Step 2 Run the **DSP NRCELL** command to verify that the value of **Cell Latest State Change Reason** is **NR Cell Setup Success**.

----End

- Perform the following steps to check whether CPRI compression has been activated and taken effect on the branch chain:

Step 1 Run the **LST NRDUCELLTRP** command to verify that the value of **Branch CPRI Compression** is **3.2:1 Compression**.

Step 2 Run the **DSP NRCELL** command to verify that the value of **Cell Latest State Change Reason** is **NR Cell Setup Success**.

----End

5.4.3 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- CPRI Specification V7.0
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *eCPRI*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- "LampSite Solution Description" in *3900 & 5900 Series Base Station Product Documentation*